ARQUITECTURAS MODERNAS DE DATOS PARA BIG DATA I/MODERN DATA ARCHITECTURES FOR BIG DATA I

Máster Universitario en Gestión / Master in Management (MIM) por la IE Universidad

Area Computer Science
Credits ECTS: 6
Term: TERM 1
Category: OPTATIVA
Language: English

SUBJECT DESCRIPTION

This course provides an overview of Big Data and the core technologies and architectures that support data-driven companies. Students will learn how Big Data solutions have enabled businesses to harness massive amounts of data to gain business insights.

By the end of the course, students will have a broad understanding of the Big Data landscape, familiarity with leading frameworks and tools, and hands-on experience applying their knowledge. This course establishes a foundation for more advanced study of data technologies, architectures and analytics. Students will be equipped to determine how Big Data can impact business needs in various domains.

This course introduces students to the principles, architectures and technologies that underpin modern Big Data ecosystems. Through a combination of conceptual discussions, hands-on laboratories and applied case studies, students will explore how organizations design scalable data platforms capable of ingesting, storing, processing and analyzing large volumes of data.

The course provides a practical and architectural overview of the Big Data lifecycle, including data ingestion, distributed storage, batch and streaming processing and large-scale analytics using technologies such as Apache NiFi, Apache Kafka and Apache Spark.

By the end of the course, students will understand the trade-offs, design considerations and operational challenges associated with modern data architectures. Students will also gain hands-on experience implementing distributed data processing workflows and using Spark APIs to analyze and transform data in scalable environments.

This course establishes the technical and conceptual foundations required for more advanced studies in distributed systems, cloud data platforms, data engineering and large-scale analytics.

LEARNING OBJECTIVES

By the end of the course, students will be able to:

- Explain the role of modern Big Data architectures in supporting scalable, data-driven organizations and business use cases.
- Compare different approaches for data ingestion, storage and distributed processing based on scalability, latency, reliability and operational requirements.
- Describe the core components and architectural principles behind technologies such as Apache NiFi, Apache Kafka and Apache Spark.
- Design simple end-to-end Big Data pipelines integrating ingestion, storage and distributed processing components.
- Apply Apache Spark APIs and DataFrames to transform, process and analyze large datasets in distributed environments.
- Evaluate the advantages and limitations of different Big Data technologies and processing paradigms for specific use cases.
- Interpret and manipulate structured and semi-structured data using Spark SQL and DataFrame APIs.
- Develop hands-on technical skills through laboratories, exploratory data analysis exercises and applied problem-solving activities.
- Relate modern data architecture concepts to broader topics in data engineering, analytics, machine learning and cloud computing.

TEACHING METHODOLOGY

This course follows a collaborative, active and applied learning approach designed to combine conceptual understanding with hands-on technical practice. Students will progressively explore the foundations of modern Big Data architectures while developing practical skills in distributed data processing technologies and analytical workflows.

The course combines lectures, guided discussions, laboratories, exploratory exercises and collaborative activities to reinforce both theoretical concepts and applied problem-solving skills. Throughout the sessions, students will analyze real-world Big Data use cases, evaluate architectural trade-offs and implement representative data processing workflows using technologies commonly adopted in industry environments.

Learning activities throughout the course include:

- Conceptual lectures introducing the principles, architectures and technologies behind modern Big Data ecosystems.
- Guided discussions and case-study analyses focused on real-world architectural decisions, scalability challenges and business-driven technology selection.
- Hands-on laboratories and technical exercises designed to reinforce practical understanding of distributed data ingestion, storage and processing systems.
- Exploratory data analysis activities using Apache Spark APIs and distributed analytical workflows.
- Knowledge checks and quizzes aimed at reinforcing key concepts and monitoring learning progression throughout the course.
- A collaborative group assignment where students will design, justify and present a representative Big Data architecture and processing workflow.

Students are expected to actively participate throughout the course, contribute to technical discussions, complete laboratory exercises and engage with the practical components of the learning experience.

Learning Activity	Weighting
Lectures	30.0 %
Discussions	10.0 %
Exercises in class, Asynchronous sessions, Field Work	30.0 %
Group work	20.0 %
Individual studying	10.0 %
TOTAL	100.0 %

In this course, we will use generative artificial intelligence (GenAI) as a tool to support your learning and exploration of Big Data concepts. However, it's essential to understand the limits and potential pitfalls of GenAI.

When using GenAI, keep the following in mind:

- If you provide minimum effort prompts, you will get low-quality results. Experiment with varying your prompt structure and specificity to achieve better results.
- Don't take GenAI's output at face value. Critically evaluate the output by asking yourself: is this answer accurate? Does it align with my knowledge and understanding of the topic?
- AI is a tool, but one that you need to acknowledge using. Failure to do so is in violation of academic honesty policies.

To acknowledge the use of GenAI, please use this format:

"I acknowledge the use of [AI systems link] to [specify how you used generative AI]. The prompts used include [list of prompts]. The output of these prompts was used to [explain how you used the outputs in your work]."

If you have chosen not to include any AI-generated content in your assignment, please ensure that this is clear and explicitly stated:

"No content generated by AI technologies has been used in this assignment."

PROGRAM

SESSION 1 (LIVE IN-PERSON)

Big Data Primer

Activities:

- Introduction to the course.
- The evolution of Data and Data Analysis.
- The origins of Big Data.
- Big Data Use Cases.
- The Big Data ecosystem.

Learning Resources:

- Session’s slides.
- Course Environment (OSBDET) introduction

SESSION 2 (LIVE IN-PERSON)

Data Ingestion

Activities:

- What’s Data Ingestion?
- Available options depending upon requirements.

Learning Resources:

- Session’s slides.
- Recordings'

SESSION 3 (LIVE IN-PERSON)

Data Ingestion - Apache NiFi

Activities:

- What’s Apache NiFi?
- Apache NiFi in the course environment.
- Alternatives to Apache NiFi

Learning Resources:

- Session’s slides.
- Some case studies of companies using Apache NiFi in production scenarios.

SESSION 4 (LIVE IN-PERSON)

Data Ingestion Lab

Activities:

- Data Ingestion
- Discussion

Learning Resources:

- Slide deck with guidelines to execute the lab.

SESSION 5 (LIVE IN-PERSON)

Data Storage

Activities:

- What’s Data Storage?
- Data Storage needs in Big Data projects.
- Available options depending upon requirements.

Learning Resources:

- Session’s slides.
- Recordings.

SESSION 6 (LIVE IN-PERSON)

Data Storage - Batch & Streaming

Activities:

- Distributed file systems and object stores - HDFS & S3
- Data Lakes
- What’s Apache Kafka?

Learning Resources:

- Session’s slides.
- Recordings.
- Some case studies of companies using HDFS & S3 in production scenarios.

SESSION 7 (LIVE IN-PERSON)

Data Storage Lab

Activities:

- Data Storage Lab
- Discussion

Learning Resources:

- Slide deck with guidelines to execute the labs.

SESSION 8 (LIVE IN-PERSON)

Data Processing

Activities:

- What is Data Processing?
- Data Processing paradigms to cope with different Use Cases.

Learning Resources:

- Session’s slides.
- Some case studies of different processing paradigms in production scenarios.

SESSION 9 (LIVE IN-PERSON)

Data Processing - Apache Hadoop

Activities:

- What is Apache Hadoop?
- YARN, HDFS, and MapReduce.
- Apache Hadoop in the Course Environment.

Learning Resources:

- Session’s slides.
- Videos related to Apache Hadoop.
- Optional MapReduce Lab.

SESSION 10 (LIVE IN-PERSON)

Data Processing - Apache Spark

Activities:

- What is Apache Spark?
- Typical Use Cases implemented with Spark.
- Apache Spark in the course environment.

Learning Resources:

- Session’s slides.
- Videos related to Apache Spark.

SESSION 11 (LIVE IN-PERSON)

Data Processing - Apache Spark, Data APIs

Activities:

- Introduction to Spark Data Abstractions.
- Spark Low-Level API.
- Spark Structured API.

Learning Resources:

- Session’s slides.

SESSION 12 (LIVE IN-PERSON)

Data Processing - Apache Spark, Data Sources API

Activities:

- Core data sources and community-driven data sources.
- DataFrames and MongoDB.

Learning Resources:

- Session’s slides.
- Notebooks with code complementing and clarifying the theoretical content.

SESSION 13 (LIVE IN-PERSON)

Data Processing - Apache Spark, DataFrames API I

Activities:

- Schemas, rows, columns and expressions.
- Conditions, types and introduction to aggregations in Spark.

Learning Resources:

- Session’s slides.
- Notebooks with code complementing and clarifying the theoretical content.

SESSION 14 (LIVE IN-PERSON)

Data Processing - Apache Spark, DataFrames API II

Activities:

- More on aggregations in Spark.
- Joins and UDFs.

Learning Resources:

- Session’s slides.
- Notebooks with code complementing and clarifying the theoretical content.

SESSION 15 (LIVE IN-PERSON)

Data Processing - Apache Spark, DataFrames API Lab

Activities:

- DataFrame API.
- Discussion.

Learning Resources:

- Slide deck and notebook with guidelines to execute the lab.

SESSION 16 (LIVE IN-PERSON)

Data Processing - Apache Spark, Reference Exploratory Data Analysis

Activities:

- Exploratory Data Analysis of Flights in the US.

Learning Resources:

- Slide deck and notebook with guidelines to execute the exercise.

SESSION 17 (LIVE IN-PERSON)

Data Processing - Apache Spark, SQL API

Activities:

- SQL Processing with Spark SQL.

Learning Resources:

- Session’s slides.
- Notebooks with code complementing and clarifying the theoretical content.

SESSION 18 (LIVE IN-PERSON)

Data Processing - Apache Spark SQL, API Lab

Activities:

- Apache Spark SQL Lab.
- Discussion.

Learning Resources:

- Slide deck and notebook with guidelines to execute the lab.

SESSION 19 (LIVE IN-PERSON)

Course Outline

Activities:

- Course review

Learning Resources:

- Session’s slides.

SESSION 20 (LIVE IN-PERSON)

Final Exam

EVALUATION CRITERIA

The evaluation system of this course is designed to assess both conceptual understanding and the practical application of modern Big Data architectures and distributed data processing technologies. Students will be evaluated through a combination of individual assessments, hands-on laboratories, collaborative work, and active class participation.

Assessment Components

- **Class Participation and Knowledge Checks (10%).** Students are expected to actively participate in classroom discussions, problem-solving activities and in-class quizzes. Participation is evaluated based on engagement, contribution to discussions and completion of knowledge-check activities throughout the course.

- **Hands-on Laboratories (10%).** Laboratories are designed to reinforce the practical understanding of Big Data technologies and distributed processing concepts. Students are expected to reproduce and execute technical exercises in the course environment using the technologies introduced during the sessions.

- **Intermediate Test (20%).** The intermediate assessment evaluates students' understanding of the foundational concepts covered during the first half of the course, including Big Data architectures, ingestion systems, storage paradigms and distributed processing fundamentals. The assessment will consist of a multiple-choice examination.

- **Final Test (30%).** The final assessment evaluates students' overall understanding of the course concepts, technologies and architectural principles. The assessment will consist of a cumulative multiple-choice examination covering both conceptual and applied topics discussed throughout the course.

- **Group Assignment - Group Deliverable & Presentation (15%).** Students will work in teams to design and present a representative Big Data use case integrating concepts and technologies covered during the course. Teams will be evaluated based on the quality of the proposed architecture, technical reasoning, clarity of presentation and alignment between business requirements and technical decisions.

- **Group Assignment - Individual Technical Walkthrough (15%).** Each student will individually explain and justify part of the technical implementation and architectural decisions associated with the group assignment. This component evaluates individual technical understanding, contribution and ability to communicate architectural concepts clearly.

INTERMEDIATE AND FINAL TESTS

In order to pass the course, **students must obtain a minimum grade of 5 out of 10 in both:**

- the Intermediate Test, and

- the Final Test.

Failure to achieve the minimum passing threshold in either examination will result in failing the course regardless of the weighted average obtained across other assessment components.

GROUP ASSIGNMENT

The group assignment will be evaluated based on:

- Technical correctness and architectural coherence.

- Alignment between business requirements and technical decisions.

- Appropriate use of Big Data technologies and distributed processing concepts.

- Clarity and quality of the presentation.
- Individual understanding and technical communication skills.

PARTICIPATION EXPECTATIONS

Students are expected to actively contribute throughout the course by participating in discussions, completing in-class knowledge checks, engaging in technical activities and collaborating during laboratory sessions.

Effective participation includes:

- Contributing constructively to class discussions and technical exercises.
- Engaging with quizzes, knowledge checks and interactive activities during sessions.
- Demonstrating preparation and involvement during hands-on laboratories.
- Collaborating respectfully and professionally with peers during group activities.
- Asking relevant technical or conceptual questions when appropriate.

Participation is evaluated based on the quality and consistency of engagement throughout the course rather than on the quantity of interventions alone.

criteria	percentage	Learning Objectives	Comments
Final Exam	30 %		40-minute multi-choice test covering the content of the whole course
Hands-on Laboratories	10 %		Hands-on Laboratories
Group Assignment - Individual Technical Walkthrough	15 %		Group Assignment - Individual Technical Walkthrough
Group Assignment - Group Deliverable & Presentation	15 %		Group Assignment - Group Deliverable & Presentation
Class Participation and Knowledge Checks	10 %		Class Participation and Knowledge Checks
Intermediate tests	20 %		Intermediate tests

FAILING GRADE AND REASSESSMENT

When students receive a Fail in a course, they have the opportunity to present themselves for reassessment in order to earn the necessary credits toward graduation.

The reassessment of students should be scheduled between 5 and 10 working days after the review session takes place.

Grades for the reassessment are limited to a Low Pass and Fail.

Both, the initial Fail as well as the grade of the reassessment remain on the transcript. For the purpose of calculating the GPA however, only the grade of the reassessment is to be considered. Students receiving a failing grade in the reassessment of a course will not be able to continue in the program.

BIBLIOGRAPHY

Recommended

- Balamurugan Balusamy; Nandhini Abirami R; Seifedine Kadry; Amir H. Gandomi.. (2021). *Big Data: Concepts, Technology, and Architecture..* 1st Edition. John Wiley & Sons, Inc.. ISBN 1119701821 (Digital) Big Data: Concepts, Technology and Architecture delivers a comprehensive treatment of Big Data tools, terminology and technology perfectly suited to a wide range of business professionals, academic researchers and students. Beginning with a fulsome overview of what we mean when we say, “Big Data,” the book moves on to discuss every stage of the lifecycle of Big Data

- Jules S. Damji, Brooke Wenig, Tathagata Das, Denny Lee. (2020). *Learning Spark.* 2nd. O'Reilly Media. ISBN 9781492050049 (Digital) Data is bigger, arrives faster and comes in a variety of formats and it all needs to be processed at scale for analytics or machine learning. But how can you process such varied workloads efficiently? Enter Apache Spark. Updated to include Spark 3.0, this second edition shows data engineers and data scientists why structure and unification in Spark matters. Specifically, this book explains how to perform simple and complex data analytics and employ machine learning algorithms

- Bill Chambers, Matei Zaharia. (2018). *Spark: the Definitive Guide.* O'Reilly. ISBN 9781491912300 (Digital) Learn how to use, deploy and maintain Apache Spark with this comprehensive guide, written by the creators of the open-source cluster-computing framework. With an emphasis on improvements and new features in Spark 2.0, authors Bill Chambers and Matei Zaharia break down Spark topics into distinct sections, each with unique goals

SOFTWARE

- *Virtual Machine.* VIRTUAL-MACHINE.

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